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Seasonal influences on preferences for urban blue-green spaces: Integrating land surface temperature into the assessment of cultural ecosystem service value

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ABSTRACT

Rapid urbanization has triggered conflicts between blue–green space and other city infrastructure, challenging urban planning efforts to improve city residents' well-being. Cultural ecosystem services (CESs) improve mental and physical wellness by offering access to blue–green space via aesthetic or recreational experiences, but these offerings fluctuate; seasonal differences and contributing factors are unclear. We divided urban green space representation into Changfeng Park (CF) in the city core district, Miracle Park (QJ) in the nearby suburb, and Wusong Paotaiwan Wetland Park (WPW) in the far suburb. We combined social media data regarding CESs with the SolVES model and innovatively considered land surface temperature (LST) as an environmental factor affecting CESs. The study demonstrated that (1) seasonal changes significantly influence CES provision in WPW and QJ via landscape dynamics and landscape-related cultural activities. In the more universally designed CF Park, CES seasonal variations were smaller. (2) LST affected CES experiences at both spatial and temporal scales, with higher degrees of influence in summer and in the WPW, potentially due to Shanghai's seasonal high temperatures and humidity and unique landscape features, respectively. (3) These findings can help create seasonally-adaptable blue–green areas that satisfy the demands of urban residents.

1. Introduction

In its latest State of the World's Cities 2022 report, UN-Habitat (2022) stated that urbanization is an unstoppable megatrend in the 21st century, particularly in developing countries, with the percentage of people living in cities expected to increase from 56 % in 2021 to 68 % by 2050. Rapid urbanization is inevitably accompanied by dramatic changes in landscape patterns and processes, including an increase in impervious surfaces and a reduction in blue–green space (Rizwan et al., 2008), which will lead to a series of environmental problems represented by the urban heat island (UHI) effect. Its negative effects include impaired air quality (Deilami et al., 2018), increased energy and water use (Kolokotroni et al., 2012), and damage to the health of urban residents and urban sustainability (Shahmohamadi et al., 2011), posing a major obstacle to urban planning endeavours designed to improve the well-being of urban residents.

Urban blue–green space combines greenery (green space) with water bodies (blue space) in the urban environment (Ma, 2023) and can be specifically generalized as blue–green infrastructure (BGI), represented by parks, gardens, etc. (Lehnert et al., 2021), which offer a variety of ecosystem services (ES) within cities. Blue–green space can benefit humans by regulating air quality and temperature extremes to lessen the negative effects of climate change (Vieira et al., 2018); intercepting rainwater to relieve pressure on urban drainage systems (Voskamp & Van de Ven, 2015); and providing habitats for a variety of plants and animals to offset biodiversity loss in the urban environment (Wen et al., 2020). Therefore, carrying out a sound assessment of blue–green space ES is essential for achieving sustainable development of the overall urban environment.

Most current assessments of urban blue–green space ES focus on local

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temperature regulation (36 %) and rainwater interception (14 %) (Veerkamp et al., 2021), but the cultural ecosystem services (CESs) that blue–green space might offer are accorded less consideration. Russell et al. (2013) defined blue–green space CESs as "ecological benefits for human well-being that are mediated by intangible processes (e.g., ideas or culture)." They enhance the physical and mental health and well-being of people who come into touch with the environment through recreational or aesthetic activities by giving them a more direct sense of nature (blue–green space) (White et al., 2020). Compared to other ES, blue–green space CESs are experienced without barriers (Andersson et al., 2015); they are more likely to be directly felt and valued by humans; thus, individual experiences with blue–green space CESs are highly subjective, which poses a challenge for the evaluation of blue–green space CESs.

Furthermore, another challenging aspect of assessing CESs in blue–green space is that CES provision patterns and people's preferences fluctuate and may differ in both temporal and spatial dimensions (Blicharska et al., 2017). Previously, Huang et al. (2023) and Wang et al. (2021) focused on the inter-annual dynamics of CESs in blue–green spaces, allowing for the assessment of CESs from the perspective of seasonal variations. An understanding of seasonal preferences for blue–green space CES value can help provide greater equality in blue–green space, thus increasing the happiness and well-being of city dwellers year-round. On a spatial scale, scholars have mostly used public participatory mapping to visualize the spatial distribution of CES value at a regional level (Xu et al., 2020), but they are often confined to studies of a single location (Clemente et al., 2019; Kong & Sarmiento, 2022) and seldom compare the spatial distribution of interregional CES values on a more macro scale.

CESs include both the unidirectional flow function that humans receive from natural ecosystems and the interactions between people and their surroundings (Bing et al., 2021). Since environmental variability affects the geographical distribution of CES values, it is critical to identify significant explanatory environmental factors and the amount to which they are influenced (Arslan & Örücü, 2021). SolVES is a model designed to determine the quantitative and spatial distribution of nonmonetary value of CESs (e.g., aesthetics and recreation) (Sherrouse et al., 2022); its embedded Maxent program was initially created to simulate the geographical distribution of species (Radosavljevic & Anderson, 2014). However, it has since been applied to a wide range of spatial modelling applications. (Chen et al., 2017). Within the context of CES assessments, Maxent software is a numerical tool that combines CES occurrences with explanatory environmental variables; it enables assessments of the contribution and importance of each environmental variable to the model's goodness of fit (Richards & Friess, 2015). In terms of the environmental variables relevant for assessing CESs in blue–green space, distance to water (DTW), distance to roads (DTR), and distance to attractions (DTA) are among the more common explanatory variables chosen (He et al., 2019; Yue et al., 2007); few studies have included land surface temperature (LST) in the assessment framework. Indeed, LST has a tremendous impact on the thermal environment of cities and people's comfort (Imran et al., 2022), which in turn affects the use of blue–green space by urban residents. Higher thermal conditions visits by urban residents to blue–green spaces (Lin et al., 2012); furthermore, where visitors congregate within blue–green spaces is strongly correlated with the thermal environment (Lin et al., 2013). Therefore, it is necessary to consider LST as one of the explanatory environmental variables for CES assessments in blue–green space.

This study applies the SolVES model to explore seasonal variations in CES values among different urban thermal environment parks in Shanghai. It inputs CES point-of-presence data obtained from social media into Maxent, to assess the impact of environmental variables (including LST, DTA, DTR, and DTW) on the spatial distribution of CES values. This research specifically attempts to respond to the following research queries: (1) How do seasonal changes affect residents' preferences for park CES values in different urban thermal environments in Shanghai? (2) Which explanatory environmental variables affect residents' preferences for CES value? In particular, does surface temperature affect visitors' perceived cultural services in parks? If so, to what extent? (3) How can the study's findings be applied to urban blue–green space landscapes and planning to adapt them to seasonal changes?

2. Method

2.1. Study area

As the distance from the city centre affects the change in surface temperature (Peng et al., 2012), within the study area, we screened the core district, nearby suburb, and far suburb of Shanghai. After considering the distance, park characteristics, and the breadth of data availability, we used Changfeng Park (CF) in Shanghai's core district, Miracle Park (QJ) in the nearby suburb, and Wusong Paotaiwan Wetland Park (WPW) in the far suburb (Fig. 1), to characterize the parks located in different urban thermal environments.

CF Park is located in the Putuo District of Shanghai, covering an area of 36.6 hectares, of which the water area accounts for approximately 40 %, and is a rare, free, large-scale comprehensive park in the city center. The park was officially opened in 1959, and it inherited a design concept of traditional Chinese garden art, drawing on the landscaping techniques of Suzhou Gardens. The combination of landscape and garden attractions is the greatest highlight of CF Park.

QJ Park is located in Minhang District, a suburb of Shanghai, with a total area of 43 hectares. It is a commercial park integrating the world's floral art, intelligent parent-child development science popularization, and cognitive education, and is known as the "Disney" of floral art parks. The park was constructed in 2016, and trial operations began in 2017. The opening of QJ Park has had a substantial impact on tourism; it has been rated as a must-do attraction in Shanghai for many consecutive years.

WPW Park is located in Baoshan District, a far suburb of Shanghai, at the convergence of the Huangpu River, the Yangtze River, and the East China Sea, with a size of 53.46 ha overall; it includes 11 ha of wetland landscapes and approximately 2250 m of shoreline along the river bank. The park, which is the largest wetland park in Shanghai and can offer reasonably rich CESs, was formally established in 2007 and is separated into a forest experience area, ecological corridor region, desert landscape area, and wetland landscape area.

2.2. Data source

2.2.1. Survey data

Meituan Dianping (https://www.dianping.com/), the first third-party review platform established in China, has pioneered the concept of independent reviews by tourists. Tourists can independently upload photos and share their experiences on the platform after their trips. To eliminate the effect of the COVID-19 outbreak on park visits in early 2020 (Tian et al., 2023), we used reviewed photos from CF Park, QJ Park, and WPW Park in 2019 as research data. Mislocated photos (e.g., located indoors or outside the park) and images featuring humans (such as selfies) or animals as the primary topic were not included in the analysis (Clemente et al., 2019). After screening out irrelevant photos, we grouped the remaining images via Table 1, which provides detailed descriptions and examples of CES value types; this information can help researchers improve classification reliability. The longitude and latitude coordinates of the photographs were determined by researchers, who carried GPS locators during field visits to the park. Finally, the seasonal classification was based on when the photos were published: spring (March-May), summer (June-August), autumn (September-November), and winter (December-February).

2.2.2. Remote sensing data

The remote sensing image data were Landsat 8–9 OLI/TIRS C2 level-

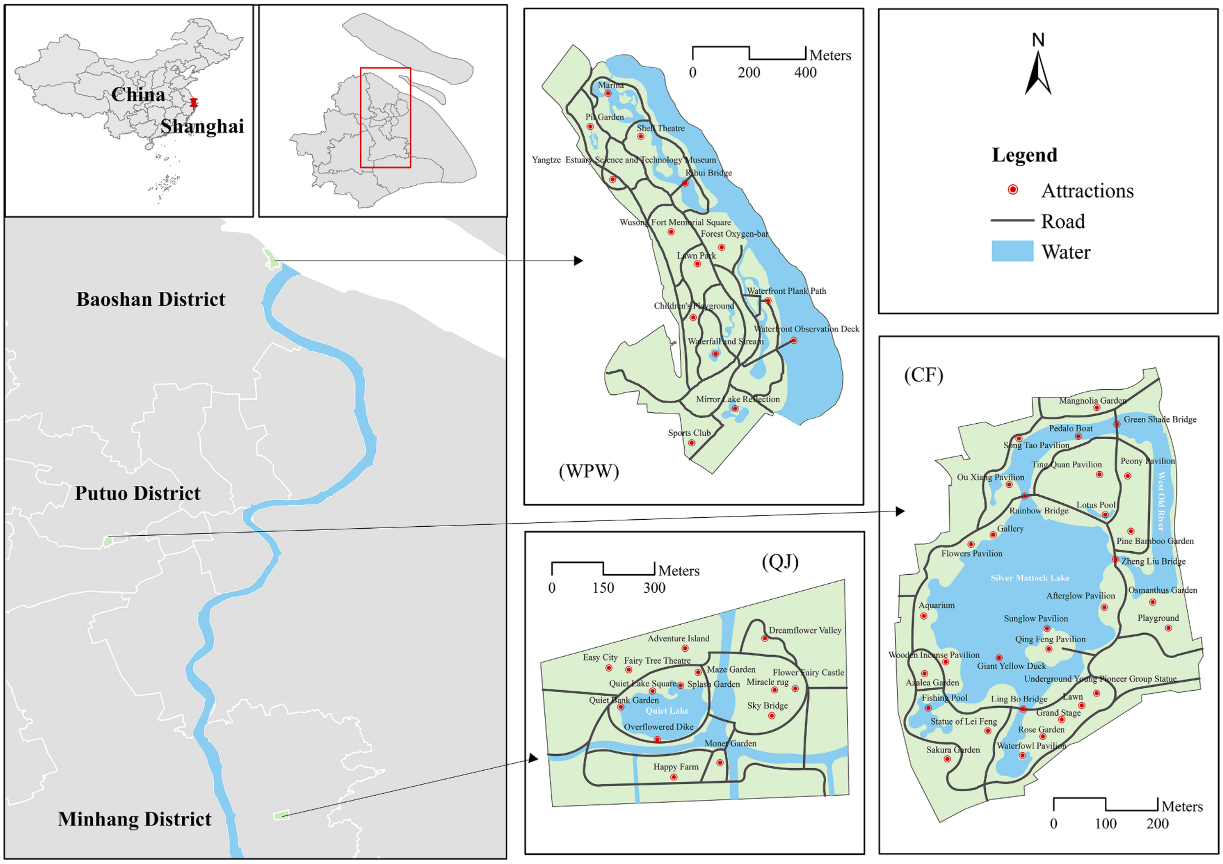

Fig. 1. Map of the study area, including CF Park, QJ Park, and WPW Park.

Table 1
Explanations and examples of CES value types used in this paper.

CES value type	Description	Examples
Aesthetic value	City parks provide rich natural and humanistic landscapes to meet people's aesthetic needs	• Viewing of landscapes, flora and fauna, and other natural, attractive characteristics • Admiration of beautiful architecture
Recreational value	City parks provide specific places for recreation and satisfy people's desire for tourism and recreation	• Relaxed outdoor recreation (e.g., walking, crabbing, boating, fishing) • Artistic creation in the environment (e.g., painting, taking personal portrait photographs)

1 data obtained from USGS Earth Explorer (https://earthexplorer.usgs.gov/); the WRS strip numbers path118 and row38 were selected for Shanghai, China. We chose remote sensing images of the park taken in clear weather with no cloud cover to minimize the impact of weather conditions on the results (Mackey et al., 2012). The screened remote sensing images were taken on 2019–5–10, 2019–7–29, 2022–10–01, and 2019–12–04, with a total of 4 views (Table 2). Notably, the remote sensing image taken on 2022–10–01 was selected as a substitute because no remote sensing image meeting the criteria was found in the autumn of 2019 (September-November), and none was available in the adjacent years.

Table 2
Remote sensing image data used in this study.

Time	Data set name
2019–5–10	LC08_L1TP_118,038_20,190,510_20,200,829_02_T1
2019–7–29	LC08_L1TP_118,038_20,190,729_20,200,827_02_T1
2022–10–01	LC09_L1TP_118,038_20,221,001_20,230,327_02_T1
2019–12–04	LC08_L1TP_118,038_20,191,204_20,200,825_02_T1

2.3. Research method

2.3.1. Land surface temperature estimation

We used the radiative transfer equation to correct for atmospheric effects (Sun et al., 2021). The formulae for calculating surface temperatures via Landsat 8 and 9 TIRS bands10 are as follows:

$$L_\lambda = M_L \times Q_{cal} + A_L \tag{1}$$

where L_λ is the top of atmosphere radiance (W·m^{-2}·sr^{-1}·μm^{-1}); M_L is the band-specific multiplicative rescaling factor (0.0003342 for TIRS band 10); Q_{cal} is the quantized calibrated pixel value; and A_L is the band-specific additive rescaling factor (0.1 for TIRS band 10).

$$L_T = \frac{L_\lambda - L_\mu^\uparrow - \tau(1-\varepsilon)L_d^\downarrow}{\tau\varepsilon} \tag{2}$$

where L_T is the surface leaving radiance; ε is the land surface emissivity; and ε is estimated according to a method based on the normalized difference vegetation index (NDVI) (Valor & Caselles, 1996). τ is the atmosphere transmission; $L_\mu^\uparrow$ and $L_d^\downarrow$ are the atmospheric downwelling and upwelling radiance, respectively. τ, $L_\mu^\uparrow$, and $L_d^\downarrow$ can be obtained by inputting the date, time, and central latitude and longitude of the remote sensing image into the atmospheric correction parameter calculator provided by NASA (Jiménez-Muñoz et al., 2010).

$$T_B = \frac{K_2}{\ln\left(\frac{K_1}{K_2} + 1\right)} - 273.15 \tag{3}$$

where T_B is the satellite brightness temperature; K_1 and K_2 are the radiance constants; for TIRS band 10, K_1 is 774.89 ($W \cdot m^{-2} \cdot sr^{-1} \cdot \mu m^{-1}$) and K_2 is 1321.08 K; and Kelvin is converted to Celsius by subtracting 273.15 (Sobrino et al., 2004). Finally, LST is derived from the following equation:

$$LST = \frac{T_B}{1 + \left(\frac{\lambda T_B}{\rho}\right) \ln\varepsilon} \tag{4}$$

where λ is the wavelength of the emitted radiance (10.9 μm for TIRS band 10) and ρ is equal to 1.438×10^{-2} mK. The above process was performed in R3.6.3 and ENVI5.1.

2.3.2. Model settings

The SolVES model combines spatial and social survey data to enable quantification, spatial analysis, and prediction of CES values. The model was divided into three submodules (Sherrouse et al., 2014): value mapping, value transfer mapping, and the ecosystem services social-values, each of which can be operated independently. Only the first two modules were utilized, while value transfer mapping was not performed in this study. The SolVES model's embedded mean nearest-neighbour tool generated R and Z values to identify the CES value's spatial distribution pattern. The model then computed a kernel density surface by using a weighted kernel density analysis on the CES points indicated by social media images, identified the maximum weighted kernel density value as the basis for normalization (Zhang et al., 2021), and finally obtained a value index (VI) of 10, with the highest VI for the CES value being the maximum value index (M-VI).

In addition, SolVES is designed to be used in conjunction with Maxent maximum entropy modelling software to enhance the expressive performance of the SolVES model (Johnson et al., 2019). Maxent is a popular tool for simulating the geographical distribution of species throughout the world (Clemente et al., 2019), and in the framework of CES evaluation, the Maxent model may examine the link between environmental factors and where specific observation sites are located. (in this study, social media photos). Since the model requires only presence data and performs well with small sample sizes, an increasing number of researchers are applying it in the assessment of CESs (Arslan & Örücü, 2021; Yoshimura & Hiura, 2017). In this study, we trained the model with 75 % of the original presence data, and we estimated the general error of the findings with the remaining data. The process was performed ten times via cross-validation, and the average of all estimations was utilized as the final output. In addition, the effects of the variables in the prediction results can be output by the Maxent model, together with their relative importance and level of contribution (He et al., 2019).

Table 3
Environment layers needed in the SolVES model.

Environment layers	Format	Description and data source
CES points	Shapefiles	Points at which tourists perceive CES are obtained according to the method in 2.1.1
Study_Area	Shapefiles	Digitized park map that uses ArcGIS10.3 to obtain park boundaries
Distance To Attractions (DTA)	Raster	Horizontal distance between the value point and the nearest attraction
Distance To Water (DTW)	Raster	Horizontal distance between the value point and the nearest water
Distance To Roads (DTR)	Raster	Horizontal distance between the value point and the nearest road
Land Surface Temperature (LST)	Raster	Data source is shown in 2.2.2 and calculated according to method 2.3.1

Table 3 displays the environmental data and resources needed for the SolVES model to function. Among them, except for the LST layer, the remaining raster layers were based on the latest map released on the park's official website; ArcGIS 10.3 was used to achieve element vectorization and rasterization via the Euclidean distance feature of the raster tools. We conducted Pearson correlation analysis on all raster layers to prevent multicollinearity, and the findings revealed that there was no meaningful link between them (r<0.3); thus, they could be input into the model for analysis.

3. Results

3.1. M-VI and mean nearest-neighbour analysis

The findings of the mean nearest-neighbour analysis (Table 4) demonstrate that the three parks' mean nearest-neighbour ratios (R) for all seasons were less than 1, their absolute values of the standard deviation (Z) were higher, and the spatial distribution of CESs was in an aggregated pattern rather than a random distribution, indicating their suitability for further analyses. To some extent, the M-VI can represent tourist preferences, and it has a considerable positive association with visitor preferences. Aesthetic value received greater attention during different periods in the three parks, and it was relatively more difficult for visitors to perceive recreational value. In addition, there were seasonal differences in visitor preferences. In general, more CES points (N) and higher M-VI were obtained in spring and autumn, while relatively few CES points and lower M-VI were obtained in summer and winter.

3.2. Seasonal variation in CES value spatial distribution in three parks

The spatial distribution of CES values for the four seasons of CF Park is shown in Fig. 2. The aesthetic value had a significantly higher VI and the largest dispersion, covering practically all of the park except for the office building in the southeast corner. The high VI was mostly concentrated on the many park attractions, such as Magnolia Garden, Peony Pavilion, Ou Xiang Pavilion, and Fishing Pool, which were connected by roads to form a spatial distribution pattern of lines connected by points. The M-VI ranking of aesthetic value for seasonal changes (Table 4) was autumn (10) > spring (9) > summer (5) > winter (3). Since boating is a featured recreational activity of CF Park, the high VI for recreational value was found near Pedalo Boat and the docks located at Rainbow Bridge and Ling Bo Bridge, with some partial planar distributions in the water at Silver Mattock Lack and West Old River. The seasonal ranking of M-VI for recreational value was consistent with aesthetic value but with lower value (2–6) and significantly smaller distributions of recreational values in summer and winter than in other seasons.

The spatial distribution of CES values in QJ Park throughout the seasons is shown in Fig. 3. Aesthetic value was distributed in a planar pattern around the park's water bodies, with high VI concentrated in Flower Fairy Castle, Miracle Rug, and Sky Bridge in the eastern part of the park, which are the park's most typical floral tourism areas. In addition, Adventure Island and Easy City in the northern part of Quiet Lake provided high aesthetic value, and the spatial distribution and high VI concentrations of recreational value were similar to those of aesthetic value. Aesthetic and recreational value had the highest M-VI of 10 and 6 in the spring, respectively (Table 4), and were more sporadically distributed and smaller in size during the other seasons, especially summer. Notably, since only 65 CES points were identified for recreational value in the summer, which is significantly fewer than the number of points (N) in the other seasons, the output VI in the SolVES model was zero.

The spatial distribution of the CES values in WPW Park during all seasons is shown in Fig. 4. The park's west-central and circular areas surrounding the wetlands along the river had the greatest concentration of aesthetic value, and high VI was mainly found at the Pit Garden, Shell

Table 4
M-VI and mean nearest-neighbour analysis for three parks.

Season	CES value type	CF				QJ				WPW			
		N	R	Z	M-VI	N	R	Z	M-VI	N	R	Z	M-VI
Spring	Aesthetic value	643	0.57	−20.69	9	1711	0.35	−51.74	10	1882	0.34	−54.60	7
	Recreational value	354	0.63	−13.43	6	1066	0.31	−43.07	6	514	0.33	−28.89	2
Summer	Aesthetic value	393	0.63	−13.86	5	123	0.37	−13.45	1	1536	0.32	−50.97	7
	Recreational value	159	0.59	−9.86	3	65	0.38	−9.51	0	599	0.30	−32.60	2
Fall	Aesthetic value	727	0.62	−19.76	10	758	0.35	−34.25	4	2488	0.38	−59.82	10
	Recreational value	251	0.63	−11.27	5	398	0.35	−24.85	2	608	0.37	−29.89	2
Winter	Aesthetic value	273	0.64	−11.44	3	628	0.44	−26.73	3	514	0.47	−22.86	2
	Recreational value	144	0.71	−6.64	2	239	0.36	−19.01	1	176	0.43	−14.42	1

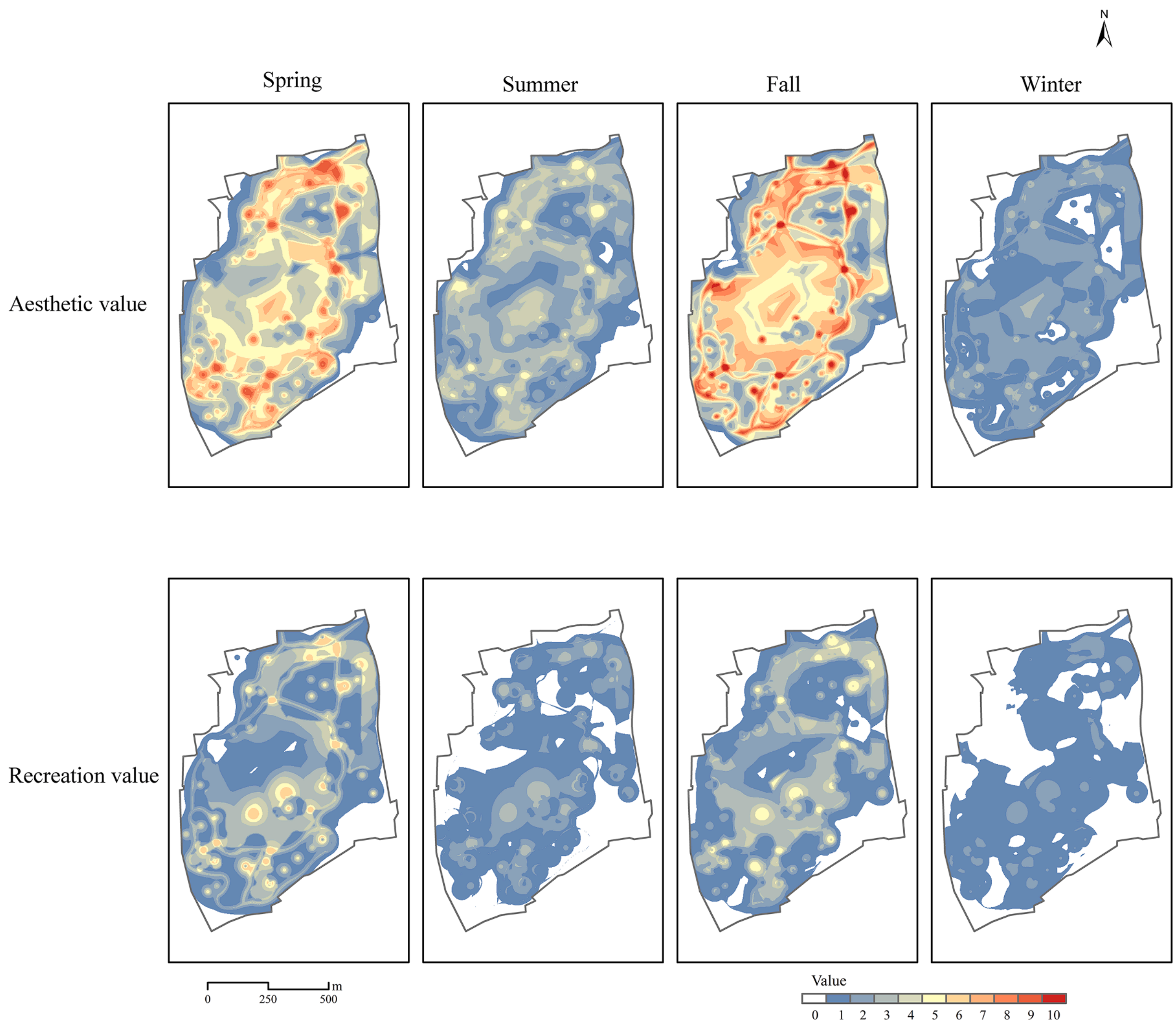

Fig. 2. Seasonal variation in CES value spatial distribution in CF Park.

Theatre, Waterfront Observation Deck, and Mirror Lake Reflection. The M-VI ranking of aesthetic value (Table 4) for seasonal changes was autumn (10) > spring (7) = summer (7) > winter (3). The recreational value was centered primarily near roadways, but the M-VI of the four seasons were lower than the aesthetic value overall (1 and 2), and the distribution area of recreational value was less dense in winter than in other seasons.

3.3. *Interseasonal variation in the spatial pattern of LST in the three parks*

Fig. 5 shows the linear fitting between the LST calculated using the method outlined in 2.3.1 and the observed air temperature at various meteorological stations within Shanghai. The results show that LST and temperature were strongly and positively connected, with a Pearson's correlation coefficient of 0.907 ($P<0.01$), and the average LST was approximately 8.96 °C warmer than the average air temperature. The agreement between LST and the observed air temperature suggests that LST can reflect the urban thermal environment.

Table 5 demonstrates the mean and standard deviation of LST for the three parks in the four seasons. The results show that in both summer and winter, the order of the mean LST values of the three parks was QJ>WPW>CF. In spring and autumn, the mean value of LST in CF Park

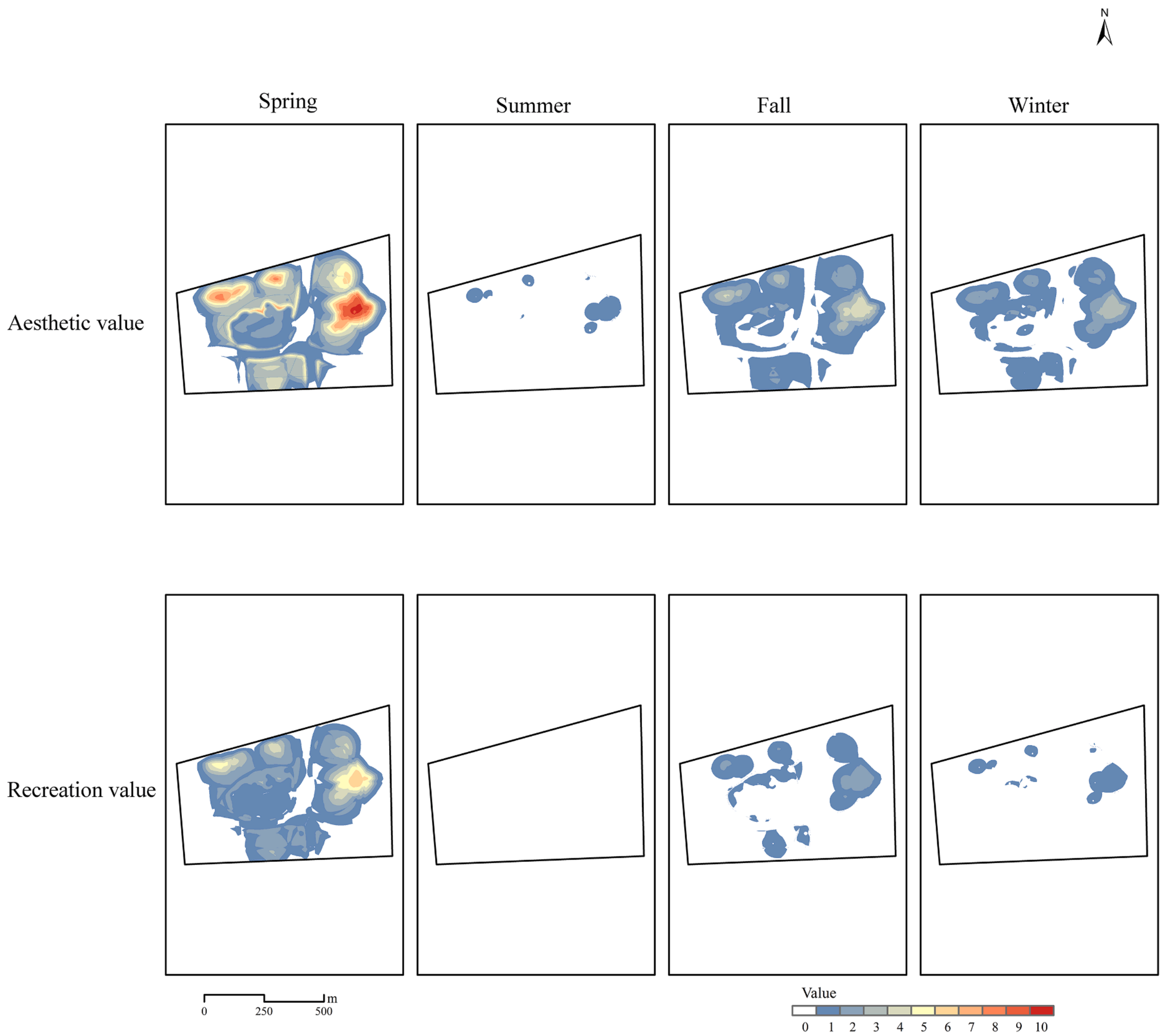

Fig. 3. Seasonal variation in CES value spatial distribution in QJ Park.

in the city center was higher than that of WPW Park in the far suburb. QJ Park, located on the outskirts of the city, had the lowest LST mean in the spring and the highest in the autumn.

Fig. 6 shows the spatial distribution of LST in the three parks during the four seasons. Overall, the spatial distribution patterns of LST were similar between seasons in all parks except for the winter season in QJ Park. The ring water area formed by the Silver Mattock Lake and West Old River in CF Park was the low-temperature zone, the park's general spatial distribution pattern revealed an upward tendency from the center to the surroundings, and the southeast and northwest corners of the park were the high-temperature areas. The low-temperature area of QJ Park was Quiet Lake in the west-central part of the park and the two rivers running through the park, while the high-temperature area was Adventure Island in the northwest corner of the park and Flower Fairy Castle in the northeast corner of the park, which were more open and dominated by human-made landscapes. The LST was lower in the wetlands around the river on the park's east side and the forested area south of the park, while the high-temperature area was located at Wusong Fort Memorial Square in the central part and Pit Garden in the northwest corner; the west and south entrances of the park also had high LSTs.

3.4. The contribution of environmental variables to CES value distribution

The spatial overlap of aesthetic and recreational values across the three parks during the four seasons is shown in Fig. 7. Pearson correlation analysis (Table 6) revealed a significant degree of correlation in the spatial distribution of aesthetic and recreational values (r>0.5). This result, together with Figs. 2–4, illustrates the fact that aesthetic and recreational values do not occur in isolation and that tourists tend to view them in combination. In addition, in conjunction with Table 4, it can be observed that the higher the M-VI of aesthetic and recreational values was, the stronger the synergy between them.

In Section 3.2, there was a more pronounced seasonal difference in the spatial distribution of CES values in the three parks. We conjectured that this result was related to the seasonal variation in LST calculated in Section 3.3. To further determine the extent of the influence of LST and other environmental variables (DTA, DTW, DTR) on the spatial distribution of CES values, we called the Maxent model independently and fed the data presented in Table 3 into the Maxent model again. The Maxent model records which variables contribute to adaptation during the training of the model. At each step of the algorithm, the model gain value is increased by correcting the coefficients corresponding to each eigenvalue, and the model assigns the increased gain value to the environmental variables on which each eigenvalue depends. At the end of the procedure, the contribution is converted into a percentage (Fig. 7). Due to the strong correlation between aesthetic and recreational values (Fig. 6), only aesthetic value points were entered here as presence data.

We found seasonal differences in the influence of LST when it was incorporated into the Maxent model as an environmental variable, with

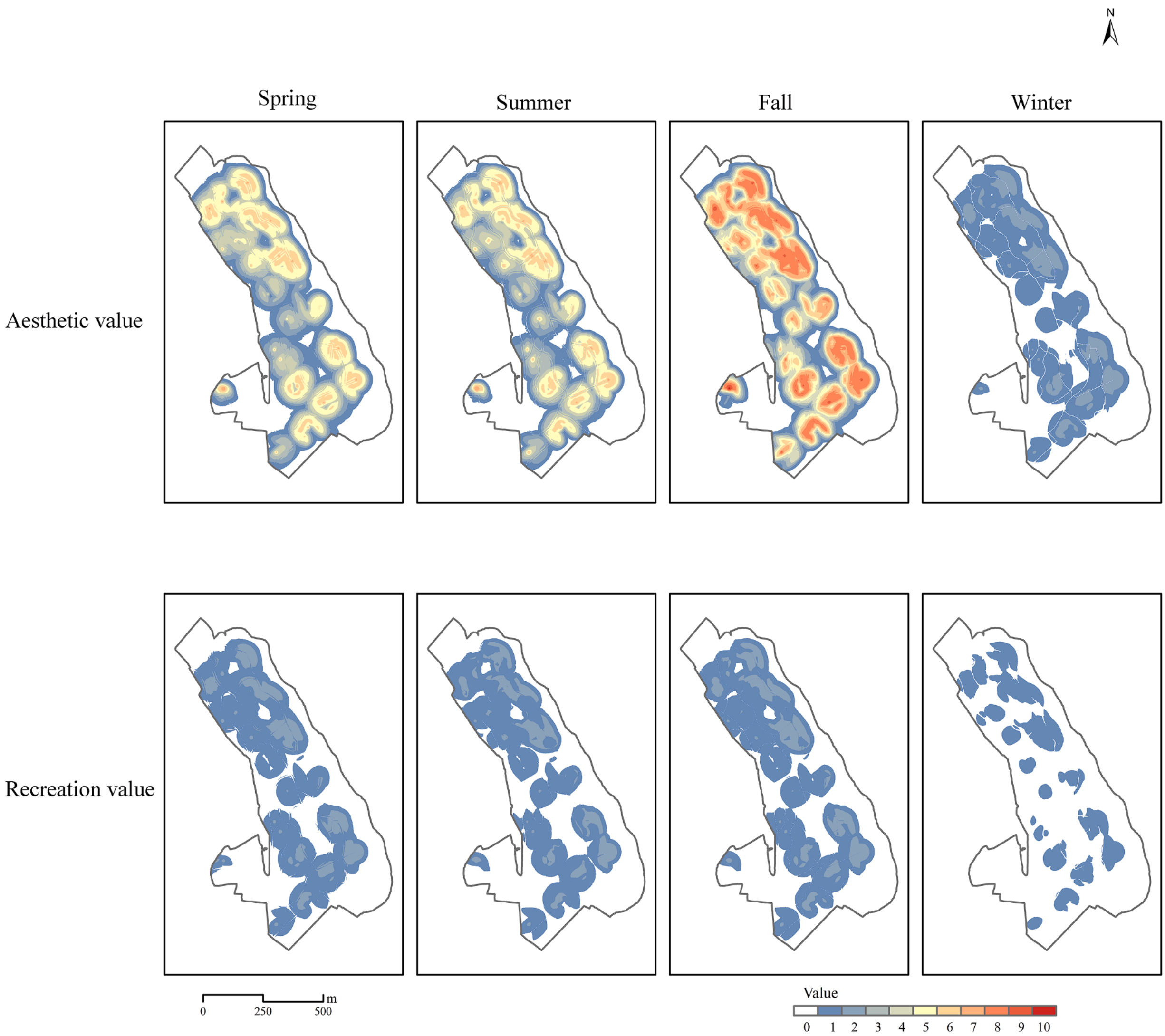

Fig. 4. Seasonal variation in CES value spatial distribution in WPW Park.

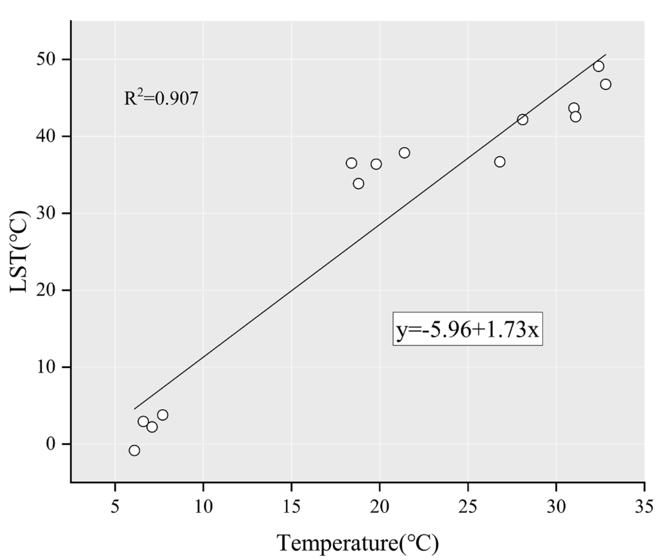

Fig. 5. Linear fitting and correlation coefficients of temperature and LST.

Table 5
Mean and standard deviation (Std) of LST for four seasons in three parks.

Season	CF		QJ		WPW	
	Mean	Std	Mean	Std	Mean	Std
Spring	26.75	2.94	24.18	1.83	25.16	2.79
Summer	36.38	2.58	38.19	2.03	37.86	2.95
Fall	32.16	2.10	33.54	1.45	30.51	1.53
Winter	−0.96	0.83	−0.15	0.62	−0.18	0.72

LST having a better fit for the model in the summer. Spatially, the LST in WPW Park, which is located in the far suburbs of the city, had a higher gain for the model. When all the environmental variables were combined, the contribution patterns of environmental variables in QJ and WPW parks were relatively similar, as shown by the fact that both DTAs contributed more than 50 % to the model, which decreased the contributions of other environmental variables, suggesting that the attractions in the two parks played a major role in determining how the CES value was distributed. The ranking of the contribution of the remaining variables was LST > DTW in WPW Park, while in the QJ garden, the order was reversed, and the contributions of DTR were low in both parks. The pattern of environmental variable contribution in CF Park was more balanced, with DTW providing the most information about the spatial

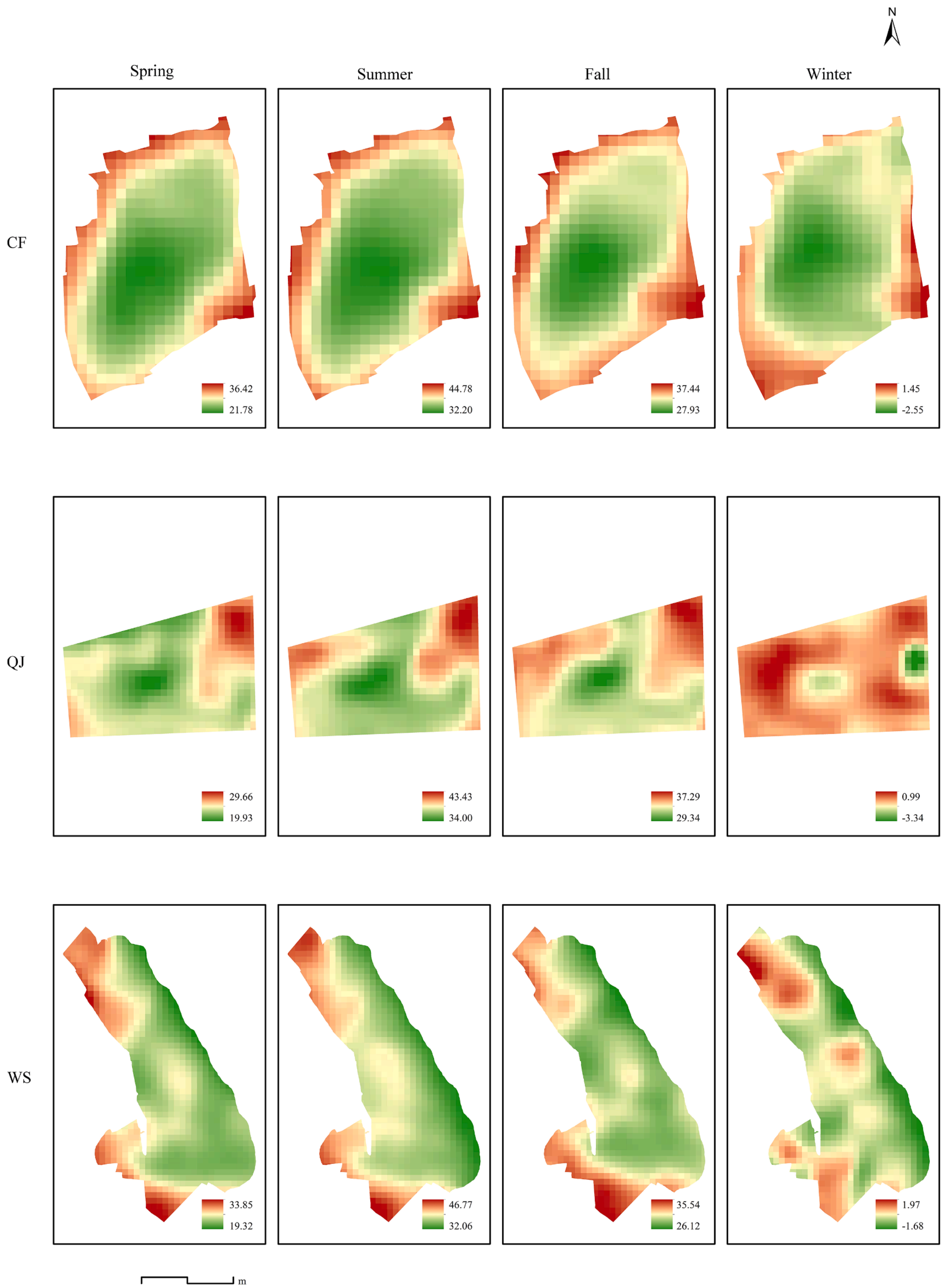

Fig. 6. Four seasons of LST (°C) in three parks.

distribution of the CES value at a level slightly higher than that of the other variables; the remaining variables contributed in the order of DTR > DTA > LST.

4. Discussion

4.1. Seasonal variations in the patterns of park CES value provision

The provision of park CESs in different urban environments located in Shanghai was much higher in both spring and autumn than in summer and winter. We speculate that meteorological differences set the stage for differences in seasonal preferences for park CES values among city

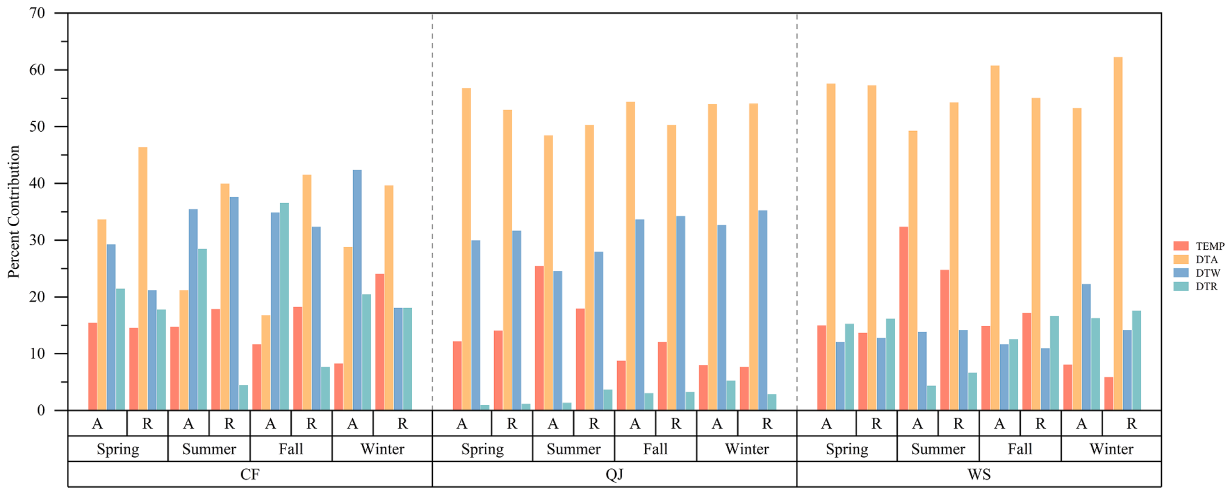

Fig. 7. Effects of environmental variables on the spatial distribution of CES values in the three parks in the four seasons.

Table 6
Correlation between CESs.

Season	CF	QJ	WPW
Spring	0.83	0.96	0.9
Summer	0.59	0	0.88
Fall	0.71	0.78	0.9
Winter	0.57	0.65	0.69

residents. Previous studies have noted that people are more likely to visit Haidian Park in Beijing when the temperature range is between 12.5 and 27.5 °C; furthermore, park visitation is also impacted by the season (Zhang et al., 2023). Hao et al. (2023) used Twitter data to conduct a study of weather on park attendance, and for every 1 °C increase in temperature during periods of high heat, the number of tweets from the parks declined by approximately 1 %. Moreover, supercooled conditions similarly impacted park visits (Hewer et al., 2016). Our study also captured fewer comments on social media platforms in summer and winter than in spring and autumn, in line with the research of the academics mentioned above.

Seasonal changes in parks impact not only their meteorological conditions but also their landscapes (Zhao et al., 2017), thus enabling urban residents to have different visual and recreational experiences in the park during the four seasons. In WPW Park, every autumn, the park is filled with flaming red maple leaves. This autumn scene has been listed by the media as one of the most beautiful autumn scenes in Shanghai, thus attracting many city residents to visit the park for photos. Autumn also has the highest seasonal CES value in WPW Park. In addition, WPW Park is located at the mouth of the Yangtze River and has a unique wetland landscape along the river, differentiating it from the other two parks; this feature provides not only aesthetic but also recreational value, such as crab catching and fishing. Furthermore, the high VI value was mostly concentrated here.

Guan et al. (2021) noted that landscape-related cultural events can further amplify the seasonal variations in park CES value offerings. QJ Park's Miracle Flower Show in the spring is a standout example of this type of event, with hundreds of flowers featured in the show. Even though there is a RMB 50 entrance fee to attend, spring has become the busiest season at QJ Park, with CES value far greater than the other three seasons. In addition, many visitors mentioned the Miracle Rug, which consists of 20,000 m^2 of flowers; as a must-see spot, the M-VI in spring was also concentrated here. The Disney-style Flower Fairy Castle in QJ Park was recognized by many parents as a good reason for residents to visit QJ Park outside of the flower season.

Seasonal fluctuations in park CES values were relatively small in parks with more universal design and landscape characteristics. CF Park is located in the center of Shanghai, with high park accessibility, making it convenient for city residents to visit the park throughout the year (Dony et al., 2015). Because the city center land resources are tight and the area of CF Park is relatively small, the park's overall landscape spatial structure is compact, open space is limited, and the attractions are contained predominantly in the gardens which have a smaller area. The spatial distribution of CES value in CF Park was also therefore presented as an obvious point distribution of the characteristics of the park. In addition, the recreational value of CF Park is higher than that of other parks, as the Silver Mattock Lake in the park can provide typical boating activities; it is a good place for visitors to have fun and thus a good place for recreation.

4.2. Influence of LST on CES values at the spatiotemporal scale

Summer in Shanghai is characterized by high temperatures and excessive humidity. According to the statistics of the Shanghai Meteorological Service, the average summer temperature in Shanghai has been ≥35 °C for 25 years. Xu et al. (2020) showed that relatively high temperatures (>35.1 °C) can negatively impact a person's body's thermal equilibrium and sensibility. Sobolewski et al. (2021) further explained the specific mechanism of this effect, in which the ability of sweat to evaporate is decreased when certain temperature and humidity thresholds are reached, thereby disrupting the body's thermal equilibrium and resulting in discomfort. In addition, both our study and (Burnett & Chen, 2021) showed consistency between LST and observed air temperatures, which explains the significant seasonal differences in the effects of LST in the study results, with LST having the greatest effect on the distribution of CES values in summer.

At the spatial scale, WPW Park in the far suburb of the city was more affected by LST. This may be related to the fact that these parks were not maintained with the same care as downtown parks. Their vegetation maintenance and management were carelessly executed, artificial structures had weakened and naturalization was obvious; furthermore, the park had a high level of vegetation cover, and general studies have suggested that vegetation cover and LST have a negative correlation (Guha & Govil, 2021; Yue et al., 2007). In addition to vegetation cover, the wetland landscape that characterizes WPW Park contributes significantly to a lower LST value. At the same time, the LST in WPW Park exhibited a larger standard deviation, indicating that the LST significantly differed between the park's hot and cool sections. As a result, tourists were more likely to travel to dense forests with high vegetation cover and cool open wetlands to experience CESs.

4.3. Value to planners

The seasonal differences in the CES values of blue–green spaces found in this study can provide a reliable reference basis for planners' decision-making.

In WPW Park, the highest threshold of its CES value occurs in autumn when the maple leaves are in full bloom, and planners should maintain and manage the corresponding landscapes to extend their viewing period. In addition, numerous commenters mentioned that they enjoy crabbing as a recreational activity in the autumn. We believe that this activity is hazardous and that planners should set up signage and fencing to provide visitors with timely warnings about the timing of high and low tides to ensure the personal safety of visitors.

For the planners of QJ Park, the surge in pedestrian flow brought about by the spring flower show often puts substantial pressure on the management of the park in the spring. Planners should efficiently manage visitation numbers and provide logistical support. In addition, the CES values of QJ Park are extremely low during other seasons, especially in summer. Planners need to design more summer facilities to increase the comfort level of visitors visiting QJ Park in summer.

Although CF Park provided more balanced CES values between seasons, it had the fewest CES points of the three parks, which may be related to its positioning as a community park. Planners need to consider offering more impactful activities or creating signature attractions to increase visitor numbers.

Notably, of the number of park CES presence points collected, the WPW park, located in the far suburb of the city, was able to provide the most CESs. The DTA of the WPW park contributed more than 50 % of the park CES in all seasons, which suggests that attractions are the most important factor in drawing tourists to these parks. This also reveals that planners should give more design features to park attractions, as Rasidi et al. (2012) show that easy accessibility does not guarantee people's visits to parks. Therefore, when park attractions are attractive enough, spatial distance and time cost will not be a deterrent for urban dwellers to visit the park.

4.4. Limitations and future prospects

Several limitations in this study can be addressed by future research. First, three parks received far more aesthetic CES points than recreational CES points (Table 4). Because the use of photographs to represent CESs injects an inherent bias that inevitably favors aesthetic value (Schirpke et al., 2018), as people tend to take more photographs of scenic locations. Furthermore, due to the overlapping nature of aesthetic and recreational value, recreational locations with less scenery tend to generate fewer photographs when people are focused on recreational activities; therefore, the recreational value of these areas tends to be underestimated.

The platform for obtaining the data for this study, Meituan Dianping, does not disclose the personal information of the commenters, and we can infer only their basic information from the pictures provided, which hinders the further depth of the study. As Dickinson and Hobbs (2017) notes, studies conducted from psychological and sociological perspectives can better analyze people's preferences for CES in urban blue–green spaces. With this method of data collection, older people who do not use social media may have been excluded from the respondents, affecting the size and audience of the data that could be collected. Further offline surveys need to be added in the future to maximize the accuracy of the CES assessment.

5. Conclusion

In this study, we applied the SolVES model to three parks in the city center (CF), near suburb (QJ), and far suburb (WPW) to assess the seasonal preferences of urban residents for the CES value of blue–green spaces in different urban thermal environments. We found that CF Park, which has a more universal design, had fewer seasonal fluctuations in CES. In contrast, WPW and QJ parks provided greater CES values in autumn and spring, respectively, due to seasonal landscape changes and landscape-related cultural activities. To further determine the effect of seasonally-driven differences in meteorological conditions on park CES, we incorporated LST into the framework of Maxent model evaluation and found that LST had a better match to the model in summer and in WPW Park, which may be related to the high temperatures and high humidity that characterize Shanghai's summers and the unique landscape features of this park.

Our study innovatively provides planners with design ideas from the perspective of seasonal changes. For example, in WPW Park, planners should strengthen the management and maintenance of the maple leaves landscape in autumn and set up relevant safety facilities. In QJ Park, planners should focus on efficiently managing pedestrian flow in spring and design more summer facilities to improve the low CES in summer. For CF Park, which has fewer CES points, planners should consider various opportunities to improve the competitiveness of the park. The results of the study can help to better satisfy residents' preferences for urban blue–green spaces in different seasons and can improve the sustained quality of urban blue–green spaces in providing CESs throughout the year.

CRediT authorship contribution statement

Sitong Huang: Conceptualization, Methodology, Writing – original draft, Formal analysis, Visualization. **Xiao Xiao:** Formal analysis, Software. **Tian Tian:** Data curation, Investigation. **Yue Che:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

Data will be made available on request.

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